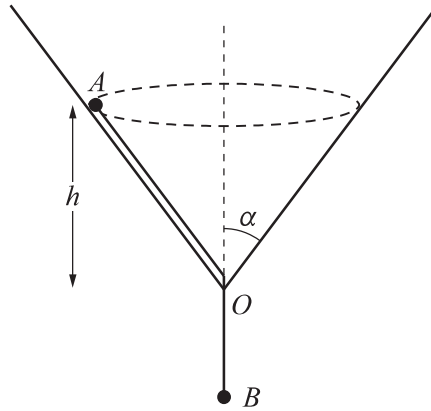


4



A hollow cone with a smooth inner surface is fixed with its vertex O downwards. The semi-vertical angle of the cone is α , where $\tan \alpha = \frac{3}{4}$. A light inextensible string has a particle A of mass m attached to one end and a particle B of mass m attached to the other end. The string passes through a small hole in the cone at O . Particle B hangs in equilibrium below O . Particle A is on the inner surface of the cone at a height h above the level of O and moves in horizontal circles with constant angular speed ω (see diagram).

Find ω in terms of g and h .

[6]

This image shows a full page of white paper with horizontal dashed lines. The lines are evenly spaced and run across the width of the page, providing a guide for handwriting practice. There are no margins, text, or other markings on the page.